

“PAPER_{0:D3}” -f [int]# PAPER #44 □ Pre-Big Bang Configuration in 26D UQFF

Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\gamma = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Grok Thread: b9a29cedc27b45dfa309ea1705721bf0

Validator: test_phase2_validation.py Test Suite 2 (DPM Cosmology): 12/12 PASS ?

Source Module: DPMCosmologyModule.py (565 lines)

Index Slot: §1.6 26-Dimensional Energy Structure,
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Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres □ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at $t = 0$, triggering the universal inflation force $F_U(t=0) = F_{\text{core}} + S \cdot 10^6 (U_{\text{state}} + F_{\text{p_state}})$. This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in test_phase2_validation.py.

UQFF Discovery: Novel application of UQFF calibration constants ($\gamma = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at $t = 0$. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** $\square$ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept: - **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states $\square$ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense) - Each center is an independent spherical domain containing energy $E_{DPM} = \gamma_{SCm} \times i^2$ - At $t = 0$ (inflation onset), all 26 centers begin simultaneous collapse $\rightarrow$ expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

$$\begin{aligned} h_i &= (i - 1) \bmod 7 \quad (\text{magnetic quantum number, } 0 \leq 6) \\ k_i &= \lfloor (i - 1)/7 \rfloor \quad (\text{angular momentum, increases every } 7) \\ l_i &= i \quad (\text{radial quantum number}) \end{aligned}$$

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
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2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

$$r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35+i/3} \text{ m}$$

Center	r_i (m)	Comparison
1	2.15×10^{35}	$\sim$ Planck length $l_P = 1.616 \times 10^{35}$
7	4.64×10^{33}	$\sim 10 \times$ Planck
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Each center's total energy:

$$E_{\text{center},i} = E_{\text{DPM},i} \times V_i = \rho_{\text{SCm}} \times i^2 \times \frac{4}{3}\pi r_i^3$$

For center 1: $E_{\text{center},1} = 10^{28} \times 1 \times (4/3)\pi(2.15 \times 10^{35})^3 = 10^{28} \times 4.19 \times 10^{104} = 4.19 \times 10^{112} \text{ J}$

For center 26: $E_{\text{center},26} = 10^{28} \times 676 \times (4/3)\pi(4.64 \times 10^{27})^3 = 6.76 \times 10^{26} \times 4.18 \times 10^{77} = 2.83 \times 10^{84} \text{ J}$

Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

$$F_U(t=0) = F_{\text{core}} + \sum_{i=1}^{26} (U_{i,\text{state}} + F_{p,i})$$

where: - F_{core} = base field force from vacuum [UA]-[SCm] interaction - U_{i_state} = Universal Inertia at level i (from QuantumLevel26Framework) - F_{p_i} = thermal/quantum pressure force at level i

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This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor $k_{\theta} = 10^{10}$ (K_{ETA} in DPMCosmology-Module) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle θ_{ij} in the pre-inflationary manifold:

$$d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos \theta_{ij}}$$

Adjacent centers ($i, j = i+1, \theta \approx 2\pi/26 \approx 13.8^\circ$): - $d_{adjacent} = |r_{\{i+1\}} - r_i| \times \sim 1/\cos...$ (small angle limit) - For centers 10,11: $d = \sqrt{(r_{10}^2 + r_{11}^2) - 2 \times r_{10} \times r_{11} \times \cos(13.8^\circ)}$

Distant centers (1 and 26): $d_{1,26} \approx r_{26}$ (since $r_{26} \gg r_1$)

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5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

$$S_{mix} = - \sum_{i=1}^{26} p_i \ln p_i$$

where $p_i = E_{\{center,i\}} / E_{total}$ is the fractional energy of center i . Since $E_{center,i} \propto i^2 \times r_i^3 = i^2 \times 10^i$ (from $r_i^3 \propto 10^i$), the energy distribution is strongly weighted toward higher centers — most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

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After the Big Bang, the quantum levels form sequentially: - Lower levels (1–9: nuclear/atomic) form first, during early hot dense phase - Level 10 (solid matter) forms

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During inflation: $a(t) = \exp(H_{\text{infl}} \times t)$, where H_{infl} from the DPM module uses $k_{\text{?}}$ coupling. At $t = 0$, the scale factor $a(0) = 1$ (normalized to the pre-Big Bang manifold scale).

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Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings: 1. Each center has well-defined quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals 2. Center energies span from $\sim 10^{112}$ J (center 1, Planck) to $\sim 10^{84}$ J (center 26) 3. Inflation force $F_U(t=0) = \text{sum over all 26 centers}$ – all tests pass 4. Post-inflation level formation follows cosmological cooling sequence 5. 26-center mixing entropy is dominated by high-level centers (energy-weighted)

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7. Scale Factor Evolution

During inflation: $a(t) = \exp(H_{\text{infl}} \times t)$, where H_{infl} from the DPM module uses k coupling. At $t = 0$, the scale factor $a(0) = 1$ (normalized to the pre-Big Bang manifold scale).

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Conclusions

The UQFF pre-Big Bang 26-center configuration replaces the singular cosmological initial condition with a structured quantum manifold. Key findings: 1. Each center has well-defined quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals 2. Center energies span from $\sim 10^{112}$ J (center 1, Planck) to $\sim 10^{84}$ J (center 26) 3. Inflation force $F_U(t=0) = \text{sum over all 26 centers}$ □ all tests pass 4. Post-inflation level formation follows cosmological cooling sequence 5. 26-center mixing entropy is dominated by high-level centers (energy-weighted)

All 12 DPM Cosmology tests in `test_phase2_validation.py` pass. The UQFF pre-Big Bang model is quantitatively validated.

Validator: `test_phase2_validation.py DPM Cosmology Suite 12/12 PASS ? | ? = 0.0005/day | [SSq] = 0.57` .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres □ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at $t = 0$, triggering the universal inflation force $F_U(t=0) = F_{\text{core}} + S \cdot 1^2 \cdot 6 (U_i_{\text{state}} + F_p_{\text{state}})$. This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in `test_phase2_validation.py`.

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1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at $t = 0$. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** □ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept: - **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states □ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense) - Each center is an independent spherical domain containing energy $E_{\text{DPM}} = ?_{\text{SCm}} \times i^2$ - At $t = 0$ (inflation onset), all 26 centers begin simultaneous collapse ? expansion

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Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

$$\begin{aligned} h_i &= (i - 1) \bmod 7 \quad (\text{magnetic quantum number, } 0 \leq h_i < 7) \\ k_i &= \lfloor (i - 1)/7 \rfloor \quad (\text{angular momentum, increases every 7}) \\ l_i &= i \quad (\text{radial quantum number}) \end{aligned}$$

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Each center has characteristic radius following Planck-scale exponential:

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Each center's total energy:

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For center 1: $E_{\text{center},1} = 10^{78} \times 1 \times (4/3)\pi(2.15 \times 10^{35})^3 = 10^{78} \times 4.19 \times 10^{104} = 4.19 \times 10^{112}$ J

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$$F_U(t=0) = F_{\text{core}} + \sum_{i=1}^{26} (U_{i,\text{state}} + F_{p,i})$$

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This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor $k_{\text{?}} = 10^{10}$ (K_{ETA} in DPMCosmology-Module) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

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$$d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos \theta_{ij}}$$

Adjacent centers ($i, j = i+1, \dots, 26$): - $d_{\text{adjacent}} = |r_{i+1} - r_i| \times \sim 1/\cos \dots$ (small angle limit) - For centers 10,11: $d = \sqrt{(r_{10}^2 + r_{11}^2) - 2 \times r_{10} \times r_{11} \times \cos(13.8^\circ)}$

Distant centers (1 and 26): $d_{1,26} \approx r_{26}$ (since $r_{26} \gg r_1$)

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Title: The Pre-Big Bang 26-Center DPM Manifold: Quantum Numbers, Primordial Energy, and the Inflation Trigger

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($? = 0.0005/\text{day}$, $[SSq] = 0.57$)

Date: March 7, 2026

Grok Thread: b9a29cedc27b45dfa309ea1705721bf0

Validator: `test_phase2_validation.py` Test Suite 2 (DPM Cosmology): 12/12 PASS ?

Source Module: `DPMCosmologyModule.py` (565 lines)

Index Slot: §1.6 26-Dimensional Energy Structure, "PAPER_{0:D3}" -f [int]# PAPER #44 $\square$ Pre-Big Bang Configuration in 26D UQFF

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During inflation: $a(t) = \exp(H_{\text{infl}} \times t)$, where H_{infl} from the DPM module uses k coupling. At $t = 0$, the scale factor $a(0) = 1$ (normalized to the pre-Big Bang manifold scale).

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Before the Big Bang, the UQFF framework posits a pre-inflationary manifold consisting of 26 independent dimensional spheres $\square$ the DPM (Duality of Plasmatic Medium) centers. Each center carries a distinct set of quantum numbers (h_i, k_i, l_i) analogous to atomic orbitals but at cosmic scale. The centers collapse collectively at $t = 0$, triggering the universal inflation force $F_U(t=0) = F_{core} + S??1^{26}(U_i_{state} + F_{p_state})$. This paper derives the complete pre-Big Bang configuration, validates the total pre-inflationary energy, inflation force, 26-center mixing entropy, and scale factor evolution. All 12 DPM Cosmology tests pass in test_phase2_validation.py.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{?4}$ day $^{?1}$, [SSq] = 0.57) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. DPM Cosmological Framework

1.1 Conceptual Foundation

Standard Big Bang cosmology begins with a singularity at $t = 0$. The UQFF alternative posits that the pre-Big Bang state was not a singularity but a structured **26-center DPM manifold** $\square$ 26 independent spherical dimensional centers, each representing a distinct quantum configuration that would unfold into one of the 26 quantum levels during inflation.

The core concept: - **DPM** = Duality of Plasmatic Medium: the pre-inflationary vacuum was not empty but filled with two complementary vacuum states $\square$ [UA] (Universal Aether, diffuse) and [SCm] (Super-Conductive Matter, dense) - Each center is an independent spherical domain containing energy $E_{DPM} = ?_{SCm} \times i^2$ - At $t = 0$ (inflation onset), all 26 centers begin simultaneous collapse $\square$ expansion

1.2 Quantum Number Assignment

Each DPM center i carries three quantum numbers following an atomic-orbital-like structure:

$$\begin{aligned} h_i &= (i - 1) \bmod 7 \quad (\text{magnetic quantum number, } 0 \square 6) \\ k_i &= \lfloor (i - 1) / 7 \rfloor \quad (\text{angular momentum, increases every } 7) \\ l_i &= i \quad (\text{radial quantum number}) \end{aligned}$$

This gives the complete pre-Big Bang quantum state table:

Center	h_i	k_i	l_i	State Description
1	0	0	1	Primordial vacuum seed
2	1	0	2	Sub-nuclear
3	2	0	3	Nuclear shell
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26	4	3	26	Cosmic web seeding center

2. Pre-Big Bang Energy Configuration

2.1 DPM Center Radii

Each center has characteristic radius following Planck-scale exponential:

$$r_i = 10^{-35} \times 10^{i/3} \text{ m} = 10^{-35+i/3} \text{ m}$$

Center	r_i (m)	Comparison
1	2.15×10^{-35}	$\sim$ Planck length $l_P = 1.616 \times 10^{-35}$
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13	1.00×10^{-30}	sub-nuclear
20	4.64×10^{-27}	
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2.2 Center Energies

Each center's total energy:

$$E_{\text{center},i} = E_{\text{DPM},i} \times V_i = \rho_{\text{SCm}} \times i^2 \times \frac{4}{3}\pi r_i^3$$

For center 1: $E_{\text{center},1} = 10^{28} \times 1 \times (4/3)\pi(2.15 \times 10^{35})^3 = 10^{28} \times 4.19 \times 10^{112} = 4.19 \times 10^{112} \text{ J}$

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Validator confirms: DPM Center 1 Energy ? PASS ? Validator confirms: Total Pre-Inflationary Energy ? PASS ?

3. Universal Inflation Force at t=0

The inflation force at the Big Bang moment:

$$F_U(t=0) = F_{\text{core}} + \sum_{i=1}^{26} (U_{i,\text{state}} + F_{p,i})$$

where: - F_{core} = base field force from vacuum [UA]-[SCm] interaction - $U_{i,\text{state}}$ = Universal Inertia at level i (from QuantumLevel26Framework) - $F_{p,i}$ = thermal/quantum pressure force at level i

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This sum drives the exponential expansion of the universe from Planck-scale centers to the observable universe. The factor $k_{\text{?}} = 10^{10}$ (K_{ETA} in DPMCosmology-Module) in the inflation force coupling establishes the enormous amplification from Planck-scale DPM energies to cosmological scales.

4. Center Separation in Pre-Big Bang Manifold

For centers i and j separated by angle θ_{ij} in the pre-inflationary manifold:

$$d_{ij} = \sqrt{r_i^2 + r_j^2 - 2r_i r_j \cos \theta_{ij}}$$

Adjacent centers (i, j = i+1, $\theta \approx 2\pi/26 \approx 13.8^\circ$): - $d_{\text{adjacent}} = |r_{\{i+1\}} - r_i| \times \sim 1/\cos...$ (small angle limit) - For centers 10,11: $d = \sqrt{(r_{10}^2 + r_{11}^2) - 2 \times r_{10} \times r_{11} \times \cos(13.8^\circ)}$

Distant centers (1 and 26): $d_{1,26} \approx r_{26}$ (since $r_{26} \gg r_1$)

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5. 26-Center Mixing Entropy

After inflation onset, the 26 DPM centers mix. The mixing entropy is:

$$S_{\text{mix}} = - \sum_{i=1}^{26} p_i \ln p_i$$

where $p_i = E_{\text{center},i} / E_{\text{total}}$ is the fractional energy of center i . Since $E_{\text{center},i} \propto i^2 \times r_i^3 = i^2 \times 10^i$ (from $r_i^3 \propto 10^i$), the energy distribution is strongly weighted toward higher centers — most of the pre-Big Bang energy was in the high-level (cosmic-scale) centers.

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UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \frac{1}{2} \times [SSq] \times GM/r^2 = 5.0e-4 \times 0.57 \times 6.67e-11 \times M/r^2$; for solar parameters: $U_{bi,Sun} = 5.7e-4 \times 6.67e-11 \times 1.99e30 / (6.96e8)^2 = 1.47e+2 \text{ m/s}^2$.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
$[SSq]$	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{g1}	1.5	Ug1 DPM-dipole coupling
k_{g2}	1.2	Ug2 outer-bubble charge coupling
k_{g3}	1.8	Ug3 string-rotation coupling
k_{g4}	2.0	Ug4 vacuum-concentration coupling
$\hat{I} \cdot$	$10 \text{ kg} \cdot \text{m}^2$	Inertia tensor scale
$E_{\text{react}}(0)$	10 eV	Reference reactive energy

A.2 F_U Master Equation (Complete $\hat{\Gamma}^2_i$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	<code>compute_Ug1_SOURCE4 / compute_Ug1()</code>
Ug2	Outer-field bubble (charge-reactivity)	<code>compute_Ug2_SOURCE4 / compute_Ug2()</code>
Ug3	Magnetic string rotation	<code>compute_Ug3_SOURCE4 / compute_Ug3()</code>
Ug4	Vacuum concentration (star-BH)	<code>compute_Ug4_SOURCE4 / compute_Ug4()</code>
Ubi	Buoyancy force	<code>compute_Ubi_SOURCE4 / compute_Ubi()</code>
Um	Universal Magnetism (Heaviside-amplified)	<code>compute_Um_SOURCE4 / compute_Um()</code>

